AEG ConnectCare™ Digital Health Platform

A Vendor-Neutral Custom Remote Patient Monitoring Platform for Hospital-to-Home Connected Care

WHITE PAPER

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Executive Summary

Remote patient monitoring (RPM) has moved beyond a pandemic-era workaround and is increasingly used to support chronic-disease management, hospital-at-home programs, specialty care, and aging in place. The attached RPM source notes that more than 80% of physicians report improved access through telehealth, six in ten U.S. adults live with a chronic disease, Medicare has established RPM reimbursement pathways, and hospital-at-home programs have federal support.

AEG and NeST Digital can use this market foundation to develop AEG–NeST ConnectedCare: a vendor-neutral hospital-to-home platform that connects multiple devices, normalizes their data, applies configurable rules and analytics, and presents clinically relevant information through a single care-team interface.

The central proposition is not to manufacture another proprietary RPM device ecosystem. Instead, AEG–NeST can create the integration, intelligence, workflow, cybersecurity, validation, and user-experience layer that allows hospitals to use appropriate connected devices without being locked into one device vendor.

The recommended starting point is a focused pilot with a limited patient population and a small set of connected devices; such as blood-pressure monitor, pulse oximeter, weight scale, thermometer and/or glucose meter; while keeping clinicians in the loop for all clinical decisions.

Goals & Objectives

Connect multiple FDA-510k cleared and consumer health devices through a secure, interoperable software layer that gives hospital care teams one operational view. AEG ConnectCare™ is a vendor-neutral connected-health platform(Figure 1) designed to integrate health data from FDA-authorized consumer and clinical-grade devices into healthcare workflows, digital patient experiences, and electronic health record (EHR) environments.

Through the AEG ConnectCare Inform API, the platform is designed to enable healthcare organizations to securely collect, normalize, organize, and deliver continuous physiologic data from multiple device manufacturers through a common integration framework. The platform is designed to support remote patient monitoring (RPM), personalized care, population-health programs, transitional care, and other digital-health initiatives while reducing the need for healthcare organizations to build and maintain separate integrations for each device manufacturer.

The AEG ConnectCare Impact solution is designed to provide clinicians and healthcare providers with a unified environment for accessing connected-device information within their clinical workflows. It is designed to provide authorized care teams with timely physiologic data from clinical-grade home-health devices to support chronic-condition management, Hospital-to-Home programs, transitions of care, patient prioritization, follow-up, and remote patient monitoring.

AEG ConnectCare is custom designed to connect with a range of in-home clinical and mobile-health devices and applications, including blood-pressure monitors, pulse oximeters, glucose monitors, ECG devices, connected scales, and other compatible health technologies through authorized APIs, SDKs, Bluetooth, Wi-Fi, and cloud interfaces.

The platform is designed around a vendor-neutral architecture to integrate and normalize health data from multiple device manufacturers into a standardized clinical-data model. It is designed to support healthcare interoperability standards such as HL7 FHIR, while incorporating appropriate privacy, security, and data-protection measures to support HIPAA-compliant healthcare environments.

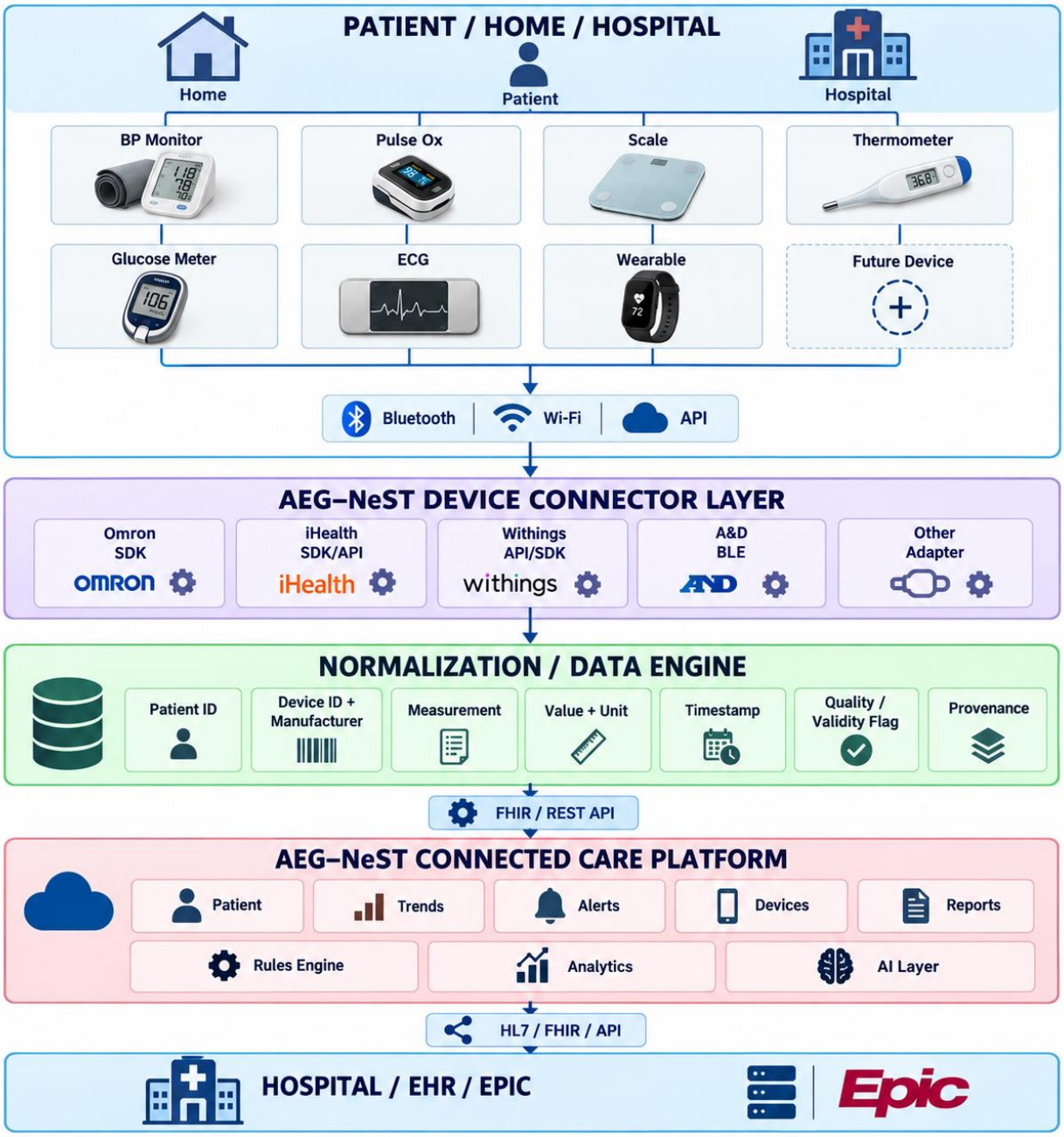


Figure 1: AEG ConnectCare™ Digital Health Platform Architecture

AEG ConnectCare™ Digital Health Platform

A Vendor-Neutral Remote Patient Monitoring(RPM) Platform for Hospital-to-Home Connected Care

I. Why RPM Matters to U.S. Health Systems

The source document frames RPM as an established and growing field driven by chronic disease, patient preference for care at home, reimbursement, and the need to extend clinical reach. It also emphasizes that RPM should extend care teams rather than replace clinicians.

RPM Driver	Evidence in Attached Source	Implication for AEG–NeST
Chronic disease	6 in 10 U.S. adults live with a chronic disease; 4 in 10 have two or more.	Large population suitable for structured monitoring between visits.
Access	80%+ of physicians report better access via telehealth.	RPM can extend hospital reach beyond the facility.
Hospital utilization	The source cites programs reporting reductions in admissions/readmissions; AdventHealth is cited with a 60% admission reduction in a high-utilizer cohort.	Pilot KPIs should include utilization and escalation metrics, while avoiding assumptions that one program's result will reproduce elsewhere.
Reimbursement	CMS maintains established RPM CPT pathways.	Workflow and documentation should be designed with reimbursement requirements in mind.
Hospital at home	The source reports 130+ CMS-approved programs by April 2024.	ConnectedCare can support hospital-to-home and transitional-care models.
Aging population	U.S. population age 65+ projected toward ~82 million by 2050.	Usability, caregiver support and low-friction connectivity are strategic requirements.

II. Proposed AEG–NeST ConnectedCare Platform

ConnectedCare would function as a software and integration layer between connected devices, patients/caregivers, hospital systems, and clinical teams. The architecture should be vendor-neutral so hospitals can select devices based on clinical need, FDA status where applicable, connectivity, cost, and patient usability.

- Device connectivity layer: cellular, Bluetooth and gateway-enabled devices.
- Interoperability layer: device APIs, standardized data model, identity matching, timestamps, units and data-quality checks.
- Rules and analytics layer: configurable thresholds, trend detection, adherence monitoring and prioritized alerts.

- Care-team workflow layer: patient roster, exception queue, escalation workflow, notes, task assignment and follow-up.
- Hospital integration layer: interfaces to EHR/EMR and other hospital applications using appropriate interoperability standards and APIs.
- Patient/Caregiver Layer: Supports patient and caregiver onboarding, reminders, education, device-status monitoring, and secure communication in accordance with applicable HIPAA privacy and security requirements.
- Security and governance layer: access control, encryption, audit logging, consent, data minimization, cybersecurity monitoring and validation.

Conceptual Data Flow

- ***CONNECTED DEVICES***
- ***SECURE CONNECTIVITY***
- ***AEG–NeST INTEGRATION LAYER***
- ***RULES / ANALYTICS / ALERTS***
- ***CARE-TEAM DASHBOARD***
- ***HOSPITAL SYSTEMS***

Human oversight remains central. The platform should surface data and exceptions to authorized clinicians; it should not independently diagnose, prescribe medication, or change therapy unless a future regulated use case is separately designed, validated, and authorized.

III. Device Strategy

The attached RPM material identifies common connected devices including thermometers, blood-pressure monitors, pulse oximeters, peak-flow meters, glucometers and scales. It also discusses wearables, environmental sensors, video monitoring and medication-management tools in the context of Alzheimer's and aging-in-place care.

Device / Data	Potential Use	Pilot Priority	Integration Note
Blood-pressure monitor	Hypertension, cardiovascular follow-up	High	Automated readings; trends and threshold alerts.
Pulse oximeter	COPD/respiratory monitoring	High	SpO_2 and pulse; escalation rules.
Weight scale	Heart failure/weight management	High	Trend detection can identify rapid changes.
Thermometer	General post-discharge monitoring	Medium	Simple vital sign for selected pathways.
Glucose meter	Diabetes monitoring	Medium	Requires carefully defined workflow and clinical ownership.
Peak-flow meter	Asthma monitoring	Pathway-specific	Useful for respiratory programs.
Wearables	Heart rate, activity, sleep	Phase 2	Higher data volume; needs signal-quality and workflow design.
Medication tools	Adherence/reminders	Phase 2	Can support adherence without automating prescribing.

Environmental sensors	Movement/routine/safety	Future specialty	Potential aging-in-place and cognitive-care applications.

V. Five-Screen Hospital Dashboard

1. Population Overview: Active patients, program status, adherence, device connectivity, open alerts, recent escalations and key operational KPIs.

2. Patient 360: Latest readings, longitudinal trends, care plan, device inventory, adherence, recent encounters and clinician notes.

3. Exception & Alert Queue: Prioritized out-of-range measurements, missing readings, device issues and configurable escalation status.

4. Workflow & Care Coordination: Tasks, ownership, outreach, escalation, follow-up, caregiver involvement and resolution tracking.

5. Program Analytics: Enrollment, engagement, alert burden, response time, utilization, readmissions/ED visits where available, and operational/reimbursement metrics.

V. Example Clinical Pathways

Hypertension

A connected blood-pressure monitor can transmit readings automatically, allowing the care team to review trends and respond to potentially dangerous spikes. AEG–NeST can add patient onboarding, adherence tracking, configurable thresholds, exception routing and documentation.

COPD / Respiratory Care

Pulse oximetry and, where appropriate, peak-flow monitoring can provide frequent measurements between visits. The source highlights the value of detecting changes that patients may not recognize themselves.

Diabetes

Connected glucose monitoring can make readings available to providers for analysis. For an initial pilot, AEG–NeST should focus on data capture, trends, workflow and clinician review rather than automated treatment decisions.

Weight / Chronic Care

Connected scales can support longitudinal weight monitoring and adherence. This can be paired with other vital signs for selected chronic-care or post-discharge pathways.

Aging in Place / Cognitive Care

The source describes wearables, environmental sensors, telehealth, video with privacy safeguards, and medication-management tools as potential components of home monitoring. This is a promising later-phase pathway after the core RPM platform is validated.

VI. AEG–NeST Roles

Workstream	AEG	NeST Digital
U.S. hospital engagement	Customer discovery, hospital relationships, use-case definition, commercialization and local program coordination.	Support technical discovery and solution demonstrations.
System architecture	Jointly define requirements, risk controls, device strategy and acceptance criteria.	Lead detailed software architecture, integration design and engineering implementation.
Software development	Product direction, U.S. requirements and validation input.	UI/UX, APIs, device integration, backend, cloud components, dashboards and test automation.
Cybersecurity / privacy	Define hospital-facing requirements, risk allocation and validation expectations.	Implement security controls, logging, identity/access management and technical evidence.
Quality / regulatory strategy	Coordinate U.S. intended-use definition and regulatory/quality review.	Engineering documentation and design controls appropriate to the agreed software classification.
Pilot execution	Hospital coordination, workflow validation, stakeholder feedback and commercialization.	Technical deployment, support, defect resolution, performance monitoring and releases.

VII. Recommended Pilot

The first pilot should demonstrate interoperability and workflow value without attempting to solve every RPM use case. A practical starting scope is 20–50 patients in one defined pathway, using two to four connected device types and one care-team workflow.

1. **Define** — Select clinical pathway; confirm intended use; map hospital workflow; choose devices; define data fields, alert logic, cybersecurity requirements, KPIs and integration boundary.
2. **Build** — Configure device connectors, patient registry, normalized data model, dashboard, alert queue, role-based access and audit logging.
3. **Integrate & Verify** — Complete interfaces needed for the pilot; test connectivity, data integrity, threshold rules, user permissions, failure modes and escalation workflow.
4. **Clinical/Operational Readiness** — Train users, enroll test patients, run simulations, validate support procedures and obtain hospital approval to start.
5. **Pilot & Evaluate** — Operate the pilot; measure adherence, connectivity, alert volume, response time, workflow burden, user satisfaction and agreed patient/utilization indicators.

VIII. Pilot Success Metrics

- ≥95% successful transmission of expected device readings after onboarding stabilization.
- Patient enrollment and activation time.
- Reading adherence and missing-reading rate.
- Alert volume per patient and proportion requiring clinical action.
- Median time from actionable alert to care-team review.
- Device/connectivity support incidents and resolution time.
- Care-team satisfaction and perceived workflow burden.

- Patient/caregiver usability and satisfaction.
- Hospital-selected clinical/utilization measures, interpreted cautiously for a small 90-day pilot.

IX. Cybersecurity, Privacy, Quality and Regulatory Boundaries

ConnectedCare would handle sensitive health information and therefore requires security and privacy to be treated as engineering requirements from the beginning. The pilot should include encryption in transit and at rest, role-based access, strong authentication, audit trails, least-privilege design, consent and data-retention rules, device identity, API security, incident response, backup/recovery, and documented verification and validation.

Regulatory classification depends on intended use and specific functionality. A platform that collects, displays, routes and supports clinician-reviewed information presents a different regulatory profile from software that independently diagnoses, recommends therapy or controls a medical device. AEG–NeST should define intended use before development and obtain appropriate regulatory/legal review rather than assume a classification.

For EHR integration, the objective should be to complement—not replace—the hospital's system of record. ConnectedCare should reduce fragmented device portals and alert burden by presenting normalized information and pushing appropriate data/workflow outputs into the hospital environment.

X. Business Model and Commercialization

The attached material supports a business case around chronic-disease burden, patient preference, clinician access, reimbursement and hospital-at-home growth. AEG–NeST can commercialize ConnectedCare as a configurable platform rather than a single-device product.

- Implementation / integration fee for initial hospital deployment.
- Per-enrolled-patient or per-active-patient platform fee.
- Enterprise license for larger health systems.
- Optional device logistics, onboarding and technical-support services through qualified partners.
- Analytics and program-performance modules.
- Custom integration and workflow-development services.

The platform should remain vendor-neutral. Device procurement can be handled directly by the hospital or through qualified device partners, while AEG–NeST focuses on the higher-value integration and intelligence layer.

XI. Risks and Mitigation

Risk	Mitigation
Alert fatigue	Start with conservative, clinician-approved rules; measure false/low-value alerts and tune thresholds.

Device fragmentation	Use an adapter/API architecture and normalized data model rather than hard-coding one vendor.
Patient technology burden	Prefer cellular or low-friction connectivity and simple onboarding.
EHR integration complexity	Keep the first pilot interface narrow; expand only after workflow value is demonstrated.
Cybersecurity/privacy	Security-by-design, auditability, penetration testing and hospital security review.
Regulatory scope creep	Lock intended use and change-control process before adding diagnostic or therapeutic functions.
Unproven ROI in a short pilot	Use operational and adoption metrics alongside longer-term clinical/utilization measures.

XII. Strategic Roadmap

- Phase 1 — ConnectedCare pilot: basic RPM device integration, dashboard, alerts and workflow.
- Phase 2 — Multi-pathway RPM: hypertension, respiratory, diabetes, weight management and post-discharge programs.
- Phase 3 — Hospital-to-Home intelligence: deeper EHR integration, analytics, patient engagement and operational dashboards.
- Phase 4 — Advanced aging-in-place / specialty monitoring: wearables, environmental sensors and caregiver workflows.
- Phase 5 — Carefully governed AI: predictive prioritization or decision-support functions only after data quality, clinical validation, regulatory strategy and human oversight are established.

XIII. Conclusion

The RPM evidence in the attached document supports a clear opportunity: hospitals increasingly need a practical way to extend care beyond the hospital while preserving clinical oversight. The differentiating opportunity for AEG–NeST is to avoid building another isolated device portal and instead create a secure, vendor-neutral interoperability and workflow platform.

AEG focused pilot can establish the technical foundation, validate hospital workflow, demonstrate device interoperability and generate the operational evidence needed for a larger U.S. commercialization program. The long-term value is a reusable ConnectedCare architecture that can expand from basic RPM into broader hospital-to-home, chronic-care and aging-in-place services.

XIV. References

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   Supporting source for Medicare RPM reimbursement, eligible services, and related billing requirements.

3. **Centers for Medicare & Medicaid Services (CMS).** *Acute Hospital Care at Home Initiative*.
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   Supporting source for physician adoption of telehealth, remote monitoring, and digital-health technologies.

5. **U.S. Census Bureau.** *The Older Population in the United States* and national population projections.
   Supporting demographic source for growth of the U.S. population age 65 and older and the corresponding need for home-based care technologies.

6. **U.S. Food and Drug Administration (FDA).** *Digital Health Technologies for Remote Data Acquisition in Clinical Investigations: Guidance for Industry, Investigators, and Other Stakeholders*.
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7. **U.S. Food and Drug Administration (FDA).** *Cybersecurity in Medical Devices: Quality System Considerations and Content of Premarket Submissions*.
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Supporting framework for cybersecurity governance, identification, protection, detection, response, and recovery.

17. **NIST.** *Implementing the Health Insurance Portability and Accountability Act (HIPAA) Security Rule: A Cybersecurity Resource Guide (NIST SP 800-66)*.
Particularly relevant to the security-by-design approach proposed for ConnectedCare.

18. **RPM Facts.** *12 Remote Patient Monitoring Statistics*, including material concerning connected health devices for chronic care and RPM applications for Alzheimer's disease and aging in place, June 2025–September 2026.

19. References to **OMRON, iHealth, Withings, A&D, Bluetooth®, Epic®, HL7®, FHIR®**, and other third-party products, technologies, standards, companies, or platforms are provided solely for **illustrative, informational, architectural, and interoperability-reference purposes**.